

hsi-wizard: A magical open-source Python package for medical hyperspectral imaging applications

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

hsi-wizard is an open-source Python package designed for processing, analysing, and visualising hyperspectral datasets, primarily in medical applications. Hyperspectral data refers to images where each pixel contains multiple intensity values at different wavelengths, with the number of spectral bands ranging from only a few to well over a thousand. These datasets vary in resolution and structure, making analysis difficult due to inconsistent formats and limited reproducibility. To address these challenges, hsi-wizard provides a standardised data representation and analysis protocols that enhance reproducibility. The package features the DataCube, a standardised class for representing hyperspectral data, which simplifies data manipulation compared to traditional array-based tools like NumPy. The DataCube includes built-in methods tailored specifically for hyperspectral data, facilitating quicker exploration and reducing complexity for users.

A notable feature of the DataCube class is its protocol functionality, which records the methods used from the hsi-wizard library. This protocol acts as a reusable template for other datasets, allowing researchers, including those without programming experience, to replicate analyses accurately and effortlessly. Additionally, hsi-wizard allows users to merge DataCubes due to its standardised representation. This capability enables researchers, for example, to easily combine near-infrared (IR) and Raman spectroscopy datasets, resulting in more comprehensive and detailed tissue analyses. Such multimodal analysis simplifies the process and provides greater insights from combined data.

Overall, hsi-wizard streamlines the workflow for processing, analysing, and visualising hyperspectral data. It caters to both beginners and experts. Students can utilise straightforward methods for educational purposes, while researchers can leverage advanced functions for professional studies, ensuring that results are reproducible and transparently documented.

Statement of Need

Hyperspectral imaging (HSI) allows for an in-depth analysis of the electromagnetic spectrum across multiple wavelengths for each pixel in an image. Initially developed for NASA applications, HSI is now applied in various fields, including medicine, agriculture, environmental monitoring and more (Bhargava et al., 2024). In medical research, for example, HSI is instrumental in distinguishing healthy tissue from cancerous areas or detecting early-stage diseases.

The evolution of these fields has led to a variety of methods for acquiring hyperspectral data, encompassing different measurement techniques such as reflection, transmission, and fluorescence, as well as varying wavelength ranges (ultraviolet, visible, infrared) and scanning methods (e.g., point scanning, line scanning, Fourier transform infrared imaging (FTIR)) (Lu & Fei, 2014). Consequently, datasets exhibit diverse formats, resolutions, and spectral ranges, which complicates analysis workflows and can hinder efficiency and reproducibility.

41 The hsi-wizard package addresses these challenges by standardising data representation
42 through its DataCube class, enabling consistent analysis across diverse hyperspectral datasets
43 and enabling data fusion across different scanning processes. Unlike traditional software tools
44 like NumPy or specialised applications such as ENVI or SPy, which often require extensive
45 programming or manual adjustments to manage variability, hsi-wizard streamlines data
46 handling and integrates multimodal datasets seamlessly. It also promotes reproducibility
47 through automated protocol logging.

48 Moreover, the extensibility of hsi-wizard allows users to incorporate new analysis methods
49 or customise existing workflows easily, ensuring adaptability to evolving research needs. This
50 flexibility guarantees that hsi-wizard remains relevant across various research scenarios,
51 significantly reducing barriers to hyperspectral data analysis while promoting transparency and
52 reproducibility in scientific research.

53 Comparison with Existing Tools

54 A variety of open-source and commercial tools exist for hyperspectral imaging, but most are
55 tailored to geospatial tasks, low-level preprocessing, or GUI-driven workflows. Libraries like
56 PySptools and SPy focus on raster I/O and classification for remote sensing, while tools such
57 as HyDe provide GPU-based denoising but lack biomedical integration. Commercial options
58 like ENVI or Spectronon offer user-friendly interfaces but are often tied to hardware and limited
59 in scripting or batch automation.

60 In contrast, hsi-wizard is designed specifically for biomedical use. It supports multimodal
61 fusion, protocol logging, and scripting within a unified Python environment. It handles formats
62 like ENVI, CSV, NRRD, and TDMS and integrates cleanly into programmatic workflows. Unlike
63 GUI-centric tools, hsi-wizard enables reproducible, automated processing for diverse spectral
64 datasets, filling a gap between algorithm libraries and rigid GUI systems.

65 This makes hsi-wizard especially suitable for clinical imaging and research, combining flexibility,
66 transparency, and domain-specific functionality in one open-source tool.

67 Example Usage

68 This snippet demonstrates the use of the hsi-wizard package for processing and visualizing
69 hyperspectral data, using a real sample from the HeiPorSPECTRAL dataset (Studier-Fischer et
70 al., 2023). The focus is on the *spleen* example (P086#2021_04_15_09_22_02). hsi-wizard
71 handles the entire pipeline: from reading the raw .dat hyperspectral DataCube, managing
72 metadata like wavelengths, to applying PCA and agglomerative spatial clustering. It enables
73 concise and structured exploration of spectral information in biomedical imaging. The full
74 example is visible in the documentation.

```
import numpy as np
import wizard
from wizard._processing.cluster import (
    pca, spatial_agglomerative_clustering, smooth_cluster
)

# Define Custom reader
def read_spectral_cube(path) -> wizard.DataCube:
    shape = np.fromfile(path, dtype=">i", count=3)
    cube = np.fromfile(path, dtype=">f", offset=12).reshape(*shape)
    cube = np.swapaxes(np.flip(cube, axis=1), 0, 1).astype(np.float32)
    wavelengths = np.linspace(500, 1000, cube.shape[2], dtype='int')
    return wizard.DataCube(
        cube.transpose(2, 0, 1),
```

```
wavelengths=wavelengths,
notation='nm',
name='HeiProSpectral',
)

# Initialize dc and read data
dc = wizard.DataCube()
dc.set_custom_reader(read_spectral_cube)
dc.custom_read('2021_04_15_09_22_02_SpecCube.dat')

# Inspect Data
wizard.plotter(dc)
```

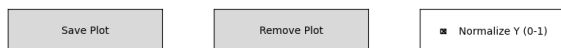
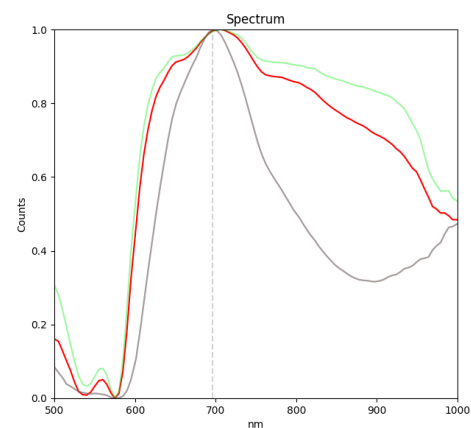
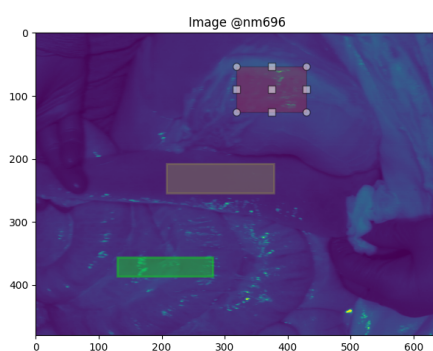


Figure 1: ROI-based spectral analysis with the interactive plotting interface of 'hsi-wizard'. The left panel shows selected tissue regions at 696 nm, while the right panel displays corresponding normalized spectra of the ROIs.

```
# Clustering
dc_pca = pca(dc, n_components=10)
agglo = spatial_agglomerative_clustering(dc_pca, n_clusters=5)
agglo = smooth_cluster(agglo, n_iter=10, sigma=0.5)
```

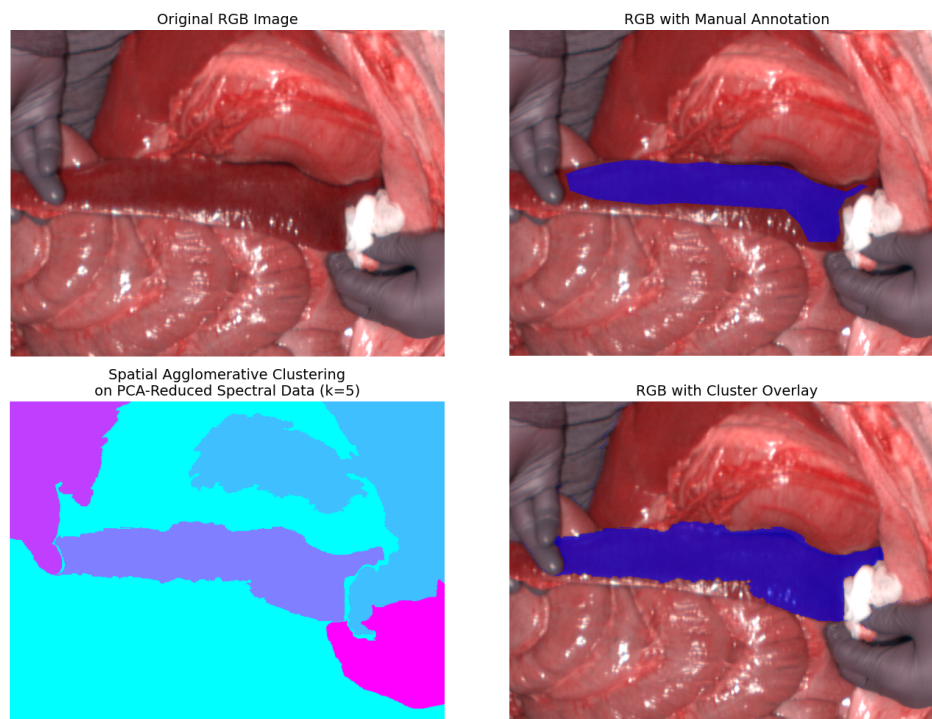


Figure 2: Comparison between manual annotation (top-right) from Studier-Fischer et al. (2023) and automated segmentation (bottom-left) using spatial agglomerative clustering ($k=5$) on PCA-reduced hyperspectral data. The original RGB image (top-left) and the resulting cluster map (bottom-right) provide visual context and output structure.

Future Work

Additional publications are currently in preparation, focusing on workflows for merging complementary datasets in more detail. The development of hsi-wizard is ongoing, with future goals that include expanding support for additional data formats and integrating advanced data analytics. We encourage feedback, contributions, and ideas from the community. All versions of hsi-wizard are available on the Python Package Index (PyPI) or on GitHub (Wuehler, 2025).

Acknowledgements

The development of hsi-wizard was funded by the CeMOS Research & Transfer Center and the Technical University of Applied Sciences Mannheim. The software was developed based on practical requirements and datasets provided by our research group: Kümmel (Kuemmel et al., 2021) shaped the mid-infrared data handling and clustering through his work on brain tissue imaging; Heintz (Heintz et al., 2021) provided multimodal Raman and VIS use cases; Manser (Manser et al., 2023) contributed workflows for Raman light sheet microscopy; Nachtmann (Nachtmann et al., 2024) validated complex 3D Raman pipelines; and van Marwick (van Marwick et al., 2024) influenced preprocessing through chemical segmentation tasks.

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